

CSMFAB000000000020 — Aegis Home Non-Conductive Motor Design: Piezoelectric and Pneumatic Motors

Version 2.0 — Revised & Expanded | June 2026

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Δ Change Log — V1.0 → V2.0

Rev	Date	Change Description
1.0	2024	Initial release — piezoelectric ultrasonic motors, pneumatic bellows actuators, PEEK gear trains, ceramic bearings, 500 N·m torque target
2.0	June 2026	Physik Instrumente (PI) 2025 ultrasonic motor data integrated (240 N·m from 200 mm diameter stator); TDK 2025 piezoelectric ceramic performance data; torque density comparison matrix (ultrasonic vs. pneumatic vs. conventional induction motor); lead-free $\text{KNbO}_3\text{-BaTiO}_3$ piezo system characterization; self-limiting torque mechanism revised; application-specific motor selection guide for domestic appliance integration

1. Executive Summary

The conventional copper-wound induction motor — the universal prime mover of domestic appliances, industrial machines, and infrastructure equipment — is a precisely engineered GIC vulnerability. It contains a ferromagnetic stator core specifically designed to concentrate and channel magnetic flux, copper windings arranged to maximize inductive coupling with that flux, and an aluminum or copper rotor casting that responds to changing magnetic fields through induction. In a Carrington-class geomagnetic storm, these design features work precisely as intended — but against the motor rather than for it. The GIC-driven magnetic flux biases the stator core into saturation, drives harmonics through the winding currents, elevates temperature, and ultimately destroys the motor through thermal runaway or insulation failure.

The Aegis Home Non-Conductive Motor Design replaces the copper-wound induction motor at the domestic appliance scale with two non-inductive alternatives: (1) piezoelectric ultrasonic motors using PZT or lead-free ceramic stacks to generate mechanical motion through elastic wave propagation rather than electromagnetic induction; and (2) pneumatic bellows actuators using compressed gas pressure in non-conductive polymer chambers. Neither

mechanism has a ferromagnetic core, neither has copper windings, and neither creates or responds to the magnetic fields that destroy conventional motors during geomagnetic storms.

Version 2.0 integrates 2025 Physik Instrumente ultrasonic motor performance data showing 240 N·m continuous torque from a 200 mm diameter stator — a threefold improvement over V1.0 projections — and full characterization of the lead-free $\text{KNbO}_3\text{-BaTiO}_3$ piezoelectric system as a direct PZT replacement.

2. Failure Analysis: Conventional Motors in GIC Events

2.1 Induction Motor Failure Mechanisms

A standard 4-pole, 3-phase induction motor (NEMA Frame 184T, 7.5 kW) contains: - Silicon steel stator laminations: 150–200 laminations, total mass 12–18 kg (ferromagnetic, $\mu_r = 4,000\text{--}8,000$ at rated flux) - Copper stator windings: 2.8–4.5 kg copper, wound in 36–48 slots - Aluminum squirrel cage rotor: 3–5 kg aluminum (or copper in high-efficiency grades)

DC Flux Bias Saturation Model:

Under GIC forcing, the geomagnetic disturbance introduces a quasi-DC flux bias (ΔB_{dc}) superimposed on the AC operating flux (B_{ac} , peak ≈ 1.5 T at rated load). The total instantaneous flux density in the stator teeth becomes:

$$B_{total}(t) = B_{ac} \cdot \sin(\omega t) + \Delta B_{dc}$$

When B_{total} exceeds the saturation flux density of the silicon steel ($B_{sat} \approx 2.0$ T), the incremental permeability drops from 4,000 to approximately 1, causing:

1. Magnetizing current increase by factor of 5–30× during the saturated half-cycle
2. Harmonic distortion — primarily 3rd harmonic (180 Hz in 60 Hz systems), amplitudes reaching 15–25% of fundamental
3. I^2R losses in stator windings increase by factor of 5–15× during saturation peaks
4. Winding temperature rise: +40–120°C above rated temperature within 5–15 minutes of GIC onset

Winding Temperature Rise Equation:

$$\Delta T = \Delta P_{loss} \times R_{th}$$

Where R_{th} (thermal resistance, winding to ambient) $\approx 0.8^\circ\text{C/W}$ for a NEMA 184T motor, and ΔP_{loss} (excess from GIC) ≈ 180 W (estimated at moderate GIC level). This yields:

$$\Delta T_{excess} = 180 \times 0.8 = 144^\circ\text{C above baseline operating temperature}$$

Class F insulation (maximum 155°C) reaches failure threshold within minutes under this loading. The motor is destroyed.

2.2 The Non-Inductive Alternative

Piezoelectric motors and pneumatic actuators have no ferromagnetic elements, no copper windings, and no aluminum rotors. The GIC failure mechanisms described above have no

pathway to affect their operation.

3. Piezoelectric Ultrasonic Motors

3.1 Operating Principle

Piezoelectric ultrasonic motors operate through the converse piezoelectric effect: an alternating voltage applied to a piezoelectric ceramic ring generates oscillating mechanical strains in the stator ring at ultrasonic frequencies (typically 35–70 kHz). The stator's elastic geometry is designed to produce an elliptical trajectory at the stator-rotor contact surface. Through friction coupling, this elliptical surface motion drives the rotor — a purely mechanical, non-electromagnetic energy conversion.

Energy conversion equations:

The piezoelectric constitutive relations relating applied field to strain are:

$$\mathbf{S} = \mathbf{s}^E \cdot \mathbf{T} + \mathbf{d}^T \cdot \mathbf{E} \quad \mathbf{D} = \mathbf{d} \cdot \mathbf{T} + \boldsymbol{\epsilon}^T \cdot \mathbf{E}$$

Where: - $\mathbf{S}$ = mechanical strain tensor - $\mathbf{T}$ = mechanical stress tensor - $\mathbf{E}$ = electric field vector
 - $\mathbf{D}$ = electric displacement - $\mathbf{s}^E$ = compliance tensor at constant $\mathbf{E}$ - $\mathbf{d}$ = piezoelectric charge constant matrix - $\boldsymbol{\epsilon}^T$ = permittivity at constant $\mathbf{T}$

For the d_{31} mode (transverse) in TDK PZT-5H (2025 specification): - $d_{31} = -274 \times 10^{-12}$ C/N (charge constant) - $d_{33} = +593 \times 10^{-12}$ C/N - $\epsilon_{33}^T/\epsilon_0 = 3,800$ - Curie temperature: 193°C - Maximum continuous operating temperature: 100°C

3.2 Physik Instrumente (PI) 2025 Motor Data (V2.0 Update)

Physik Instrumente, the leading commercial developer of piezoelectric ultrasonic motors, published in 2025 performance data for their advanced 200 mm diameter ring-type ultrasonic motor, representing the state of the art in high-torque piezoelectric motor design:

PI 2025 Ultrasonic Motor — 200 mm Diameter Specification:

Parameter	PI 2025 (200 mm)	PI Previous Gen (200 mm)	CSM V1.0
Continuous torque	240 N·m	80 N·m	120 N·m
Peak torque	480 N·m	160 N·m	240 N·m
No-load speed	180 RPM	120 RPM	100 RPM
Rated speed (at 240 N·m)	95 RPM	55 RPM	—
Electrical input power	950 W	600 W	—
Efficiency (at rated torque/speed)	72%	58%	—
Stator material	Lead-free KNN ceramic	PZT-5A	PZT-5H
Mass (motor only)	3.8 kg	4.2 kg	—
Metal content	Titanium stator backing plate	Titanium	—

Torque density comparison:

$$\text{Torque density} = \text{Peak torque} / \text{Motor mass} = 480 \text{ N·m} / 3.8 \text{ kg} = 126 \text{ N·m/kg}$$

For comparison, a conventional 7.5 kW 4-pole induction motor producing equivalent torque at rated speed has a torque density of approximately 12–18 N·m/kg at the motor shaft. The ultrasonic motor's 126 N·m/kg represents a 7–10× torque density advantage, enabling smaller, lighter motor installations.

3.3 Lead-Free KNbO₃-BaTiO₃ System (V2.0 Update)

The most significant 2025 advance in piezoelectric motor technology is the commercial availability of lead-free ceramic systems achieving near-PZT performance. The KNbO₃-BaTiO₃ (KNN-BT) system, which eliminates lead — a toxic heavy metal requiring REACH registration and RoHS compliance documentation — now achieves:

KNbO₃-BaTiO₃ (KNN-BT) vs. PZT-5H — TDK 2025 Data:

Parameter	TDK PZT-5H (2025)	TDK KNN-BT (2025)	Requirement
d ₃₃ (pC/N)	593	420	≥ 350
d ₃₁ (pC/N)	≥ 274	≥ 185	≥ ≥ 150
k _p (planar coupling)	0.63	0.52	≥ 0.45
Curie Temperature (°C)	193	310	≥ 200
Mechanical Q	65	95	—
Dielectric constant (ε ₃₃ /ε ₀)	3,800	1,850	—
Lead content	65 wt% (RoHS exempt)	Zero	Zero (Aegis target)
Density (g/cm ³)	7.8	4.6	—
Electrical resistivity	> 10 ¹⁰ Ω·m	> 10 ¹² Ω·m	> 10 ¹⁰

The KNN-BT system's **elevated Curie temperature of 310°C** — versus 193°C for PZT-5H — provides a substantially larger operating margin below the depolarization point, which is critical for motor applications where localized frictional heating at the stator-rotor contact can reach 60–80°C above ambient. The PZT-5H system at 100°C operating temperature is within 93°C of its Curie point (approximately 50% of the thermal safety margin); KNN-BT at 100°C is within 210°C of its Curie point (68% safety margin).

Aegis Home Specification: KNN-BT replaces PZT-5H in all new Aegis motor production effective Version 2.0.

3.4 Aegis Home Ultrasonic Motor — 500 N·m Configuration

For the domestic appliance scale (washing machine, dishwasher, HVAC blower — up to 500 N·m continuous torque requirement), the Aegis Home motor specification uses two 200 mm PI-equivalent KNN-BT stators operating on a common rotor:

Dual-Stator Configuration Performance:

$$T_{\text{total}} = 2 \times T_{\text{single}} \times \eta_{\text{coupling}} = 2 \times 240 \text{ N·m} \times 0.95 = 456 \text{ N·m continuous}$$

Peak torque:

$$T_{\text{peak}} = 2 \times 480 \times 0.95 = 912 \text{ N·m}$$

This exceeds the 500 N·m target from V1.0 by 82% at rated conditions, providing significant derating margin for temperature and aging effects.

Motor Assembly Specification:

Parameter	Value
Configuration	Dual-stator, single rotor
Stator diameter	200 mm (×2)
Overall motor diameter	220 mm
Overall motor length	95 mm
Total mass	8.2 kg
Continuous torque	456 N·m
Peak torque	912 N·m
Speed range	10–200 RPM (variable via drive frequency)
Drive frequency	40–65 kHz (KNN-BT resonant range)
Self-limiting mechanism	Stator-rotor friction saturation above 500 N·m (rotor slips)
Metal content	Titanium stator backing plates (×2), ceramic ball bearings
Total electrical conductor	Drive electrodes only (< 50 cm total copper track)

3.5 Self-Limiting Torque Safety Mechanism

The friction coupling between stator and rotor in an ultrasonic motor provides an inherent self-limiting torque: when the applied mechanical load exceeds the friction coupling capacity (determined by the normal contact force, contact area, and friction coefficient), the rotor slips against the stator surface. This is not a failure — it is a designed safety feature. The motor cannot transmit more than its friction-limited torque regardless of stator excitation amplitude:

$$\mathbf{T_limit} = \mu_k \times \mathbf{F_n} \times \mathbf{r_contact}$$

Where $\mu_k = 0.3$ (ceramics on PEEK rotor contact layer), $\mathbf{F_n} = 2,400$ N (stator-rotor preload), $\mathbf{r_contact} = 95$ mm (mean contact radius):

$$\mathbf{T_limit} = 0.3 \times 2,400 \times 0.095 = 68.4 \text{ N}\cdot\mathbf{m} \text{ per contact zone}$$

With 7 contact zones per stator and 2 stators:

$$\mathbf{T_limit_system} = 68.4 \times 7 \times 2 = 958 \text{ N}\cdot\mathbf{m}$$

This self-limiting mechanism eliminates the need for an external mechanical overload clutch — the motor inherently cannot transmit destructive torques, making it intrinsically safe for domestic appliance applications.

4. Pneumatic Bellows Actuators

4.1 Operating Principle

Pneumatic bellows actuators convert compressed gas pressure into mechanical force through elastic deformation of a sealed polymer bellows chamber. The force output is simply:

$$F = P_{\text{gauge}} \times A_{\text{effective}}$$

Where $A_{\text{effective}}$ is the effective cross-sectional area of the bellows convolution. For a 120 mm diameter bellows at 6 bar gauge pressure:

$$F = 600,000 \text{ Pa} \times \pi \times (0.06)^2 = 600,000 \times 0.01131 = 6,785 \text{ N} \approx 6.8 \text{ kN}$$

This force output is appropriate for linear actuator applications (linear displacement of 50–200 mm range) at the domestic appliance scale.

4.2 Torque Density Comparison — Ultrasonic vs. Pneumatic vs. Conventional

Application-Specific Motor Selection Guide (V2.0):

Application	Required Torque	Required Speed	Recommended Technology	Rationale
Washing machine drum	40–120 N·m	50–1,200 RPM	Ultrasonic motor	High speed range, compact form
HVAC blower	5–20 N·m	600–3,000 RPM	Ultrasonic motor	High speed, low torque
Dishwasher pump	2–8 N·m	1,500–3,000 RPM	Ultrasonic motor	High speed
Industrial valve actuator	100–500 N·m	0.1–5 RPM	Pneumatic bellows + ratchet	High torque, low speed, high reliability
Linear gate valve	5–50 kN force, 50–200 mm stroke	N/A (linear)	Pneumatic bellows direct	Direct linear force, no rotary conversion
Press or clamping device	10–80 kN	N/A (linear)	Pneumatic bellows	High static force, sustained hold

Torque Density Comparison Matrix:

Motor Type	Torque Density (N·m/kg)	Power Density (W/kg)	GIC Immunity	Notes
Ultrasonic (KNN-BT, 200mm)	126 N·m/kg	250 W/kg	Complete	Best domestic appliance choice
Pneumatic bellows (120mm, 6 bar)	15 N·m/kg (equivalent)	45 W/kg	Complete	Best for high-force linear applications
NEMA 184T Induction (7.5 kW)	15 N·m/kg	2,800 W/kg	None	Destroyed in G5 event
BLDC (brushless DC, equivalent)	22 N·m/kg	1,500 W/kg	Partial (no ferrite core)	Copper windings still vulnerable

The ultrasonic motor demonstrates a $5.7\times$ torque density advantage over conventional induction motors and $8.4\times$ over pneumatic bellows actuators — but the pneumatic system retains decisive advantages for sustained static force applications where the actuator is pressurized and held without consuming additional energy.

4.3 PEEK Gear Train for Speed Conversion

When ultrasonic motor speeds (10–200 RPM rated) must be converted for higher-speed applications (HVAC blowers at 1,500–3,000 RPM), PEEK gear trains provide the required step-up ratio while maintaining non-conductive, non-magnetic construction:

PEEK Gear Specification (Victrex® PEEK 450G):

Parameter	Value
Tooth form	Involute, module 1.5 mm
Material	PEEK 450G (unfilled)
Tensile strength	100 MPa
Compressive strength	118 MPa
PV limit (dry running)	0.16 MPa·m/s
Maximum pitch line velocity	8 m/s
Electrical resistivity	$> 10^{16} \Omega\cdot\text{m}$

Parameter	Value
Maximum service temperature	250°C

Gear Life Calculation (Hertz contact stress):

For a PEEK pinion ($N_1 = 20$ teeth) meshing with a PEEK wheel ($N_2 = 100$ teeth), module 1.5 mm, at 240 N·m input torque and 95 RPM:

Tangential force at pitch circle:

$$F_t = 2T / (m \times N) = 2 \times 240 / (0.0015 \times 20) = 16,000 \text{ N}$$

Contact stress:

$$\sigma_H = Z_E \times \sqrt{(F_t \times (u+1) / (b \times d_1 \times u))} = 134 \times \sqrt{(16000 \times 6 / (0.05 \times 0.030 \times 5))}$$

$$\sigma_H \approx 134 \times \sqrt{(12,800)} \approx 134 \times 113 = 15,142 \text{ MPa} \text{ — EXCEEDS PEEK capability}$$

This analysis confirms that at 240 N·m input torque, PEEK gears cannot directly transmit the full ultrasonic motor output for small-module gears. The Aegis Home system resolves this through: 1. Stage 1: Ceramic (Si_3N_4) gears for the first reduction from motor output (240 N·m $\rightarrow$ 80 N·m at 3:1 stage, module 3 mm) 2. Stage 2: PEEK gears for the final stage (80 N·m $\rightarrow$ final output, larger module 3–5 mm)

This hybrid ceramic-PEEK gear train maintains non-conductive construction while correctly distributing load across gear stages appropriate to each material's contact stress limits.

5. Ceramic Ball Bearings — Non-Sparking Specification

5.1 Ceramic Ball Bearing Selection

All bearing positions in Aegis Home motors use Si_3N_4 ceramic balls in PEEK races:

Bearing Specification — Non-Sparking Grade:

Parameter	Value
Ball material	Silicon nitride (Si_3N_4 grade G5)
Race material	PEEK 150GL30 (glass-filled, higher modulus)
Retainer material	PEEK 450G (unfilled)
Lubrication	Fluorinated grease (NLGI Grade 2, per PTFE carrier)
Electrical resistivity	$> 10^{12} \Omega$ (bearing assembly)
Spark generation	None (ceramics are non-sparking)
Operating temperature	-40°C to $+200^\circ\text{C}$

Parameter	Value
Load rating	35% below equivalent steel bearing (design compensation required)

The non-sparking property of ceramic bearings is relevant beyond the GIC scenario — in post-Carrington environments where flammable gases may be present from damaged infrastructure, non-sparking motor internals prevent secondary ignition hazards.

6. Non-Sparking, Non-Conductive Motor — Complete System Integration

6.1 Fully Assembled Motor Properties

The complete Aegis Home Non-Conductive Motor System for domestic appliance scale is summarized:

System Property	Value
Continuous torque output	456 N·m (dual stator)
Peak torque	912 N·m
Speed range	10–200 RPM (ultrasonic stage)
Power consumption	1,900 W (dual stator at rated)
Efficiency	72% (motor stage)
Total ferromagnetic content	Zero
Total copper content	< 50 cm electrode track
Electrical resistivity (housing)	> 10^{14} $\Omega\cdot\text{m}$ (PEEK/ceramic)
Spark generation	None
GIC coupling coefficient	Zero
Self-limiting torque	Yes (friction slip at 958 N·m)
Motor housing material	PEEK 450G with ZrO ₂ face plates
Operating environment	IP65 (dust-tight, water jet protected)
Certification target	IEC 60034-18 (rotating electrical machines — adapted for non-electrical motor)

7. Fabrication Protocol

KNN-BT Piezoelectric Ceramic Procurement (TDK 2025 spec)

□
 Ring Stator Machining (ID/OD grinding, parallel face grinding, $R_a \leq 0.2 \mu\text{m}$)
 □
 Electrode Screen Printing (silver paste, fired at 850°C , 30 min, air)
 □
 Stator-Backing Plate Bonding (titanium backing, Araldite 2031 film adhesive)
 □
 Poling: 40 kV/cm DC field, 120°C , 30 min (KNN-BT poling protocol)
 □
 Frequency Sweep Characterization (impedance analyzer, verify resonant frequency $\pm 200 \text{ Hz}$)
 □
 PEEK Rotor Machining (contact surface $R_a \leq 0.1 \mu\text{m}$, flatness $\leq 2 \mu\text{m}$)
 □
 Si $\square$ N $\square$ /PEEK Gear Train Assembly (ceramic first stage, PEEK second stage)
 □
 Bearing Assembly (Si $\square$ N $\square$ balls, PEEK races, fluorinated grease)
 □
 Motor Assembly in PEEK Housing (controlled torque fasteners, ceramic lock washers)
 □
 Break-in Run (10 cycles, full stroke, unloaded)
 □
 Torque Calibration (dynamometer, verify 456 N·m continuous, 912 N·m peak)
 □
 Dielectric Withstand Test (10 kV applied to housing vs. rotor, 1 min)
 □
 GIC Immunity Verification (50 mT DC bias applied, verify zero torque change)

8. Quality Acceptance Criteria

Parameter	Specification	Test Method
Continuous torque	□ 456 N·m (dual stator)	Dynamometer
Speed range	10–200 RPM $\pm$ 2%	Tachometer
Resonant frequency (each stator)	Within $\pm 200 \text{ Hz}$ of target	Impedance analyzer
Motor efficiency	□ 68% at rated load	Input/output power measurement
Electrical resistivity (housing)	$> 10^{12} \Omega$	ASTM D257
Dielectric withstand	10 kV, 1 min, no breakdown	IEC 60243-1
GIC immunity (50 mT bias)	$< 2\%$ torque variation	Electromagnet + dynamometer
Bearing temperature rise	$< 35^\circ\text{C}$ at rated load	Thermocouple
Self-limiting engagement	Slip at 900–1,000 N·m	Controlled load test

Parameter	Specification	Test Method
Curie temperature (stator ceramic)	□ 300°C	DSC thermal analysis

9. References and Standards

- Physik Instrumente (PI) GmbH, Ultrasonic Piezo Motor Catalog 2025
- TDK Corporation, Piezoelectric Ceramics Product Specifications 2025 (PZT and KNN-BT series)
- IEC 60034-18: Rotating Electrical Machines — Functional Evaluation of Insulation Systems
- ASTM D257: Standard Test Methods for DC Resistance or Conductance of Insulating Materials
- IEC 60243-1: Electric Strength of Insulating Materials
- Saito, Y. et al., “Lead-free Piezoceramics,” *Nature*, Vol. 432, 2004; extended commercial performance data TDK 2025
- “Geomagnetic Storm Effects on AC Induction Motor Performance,” *IEEE Transactions on Industry Applications*, Vol. 61, No. 2, 2025
- Victrex PEEK 450G Engineering Properties, Victrex plc, 2024

“A conventional motor is a geomagnetic antenna with mechanical ambitions. The Aegis Non-Conductive Motor has no ferromagnetic core to saturate, no windings to heat, and no inductive geometry to offer to the storm. It operates by elastic strain, not electromagnetic flux — and the storm has nothing to say about that.”

— **CSM Engineering Design Directive, Aegis Home Series**

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